

- 1] For the process $X(t) = 2\cos(t + \theta)$ where $\theta \sim U(0, \pi)$, we have $E(X^2(t)) =$
 A) 5π ☒ B) π C) 2π D) 3π E) None
- 2] The variance of a mean-ergodic process $X(t)$ with no periodic component and $R_X(\tau) = \pi + \pi^2 e^{-|\tau|}$ is equal to
☒ A) π^2 B) π C) 2π D) 3π E) None
- 3] For a random process $X(t) = 2\cos(\phi)$ where $\phi \sim U(0, \pi)$, we have $P_{X^2} =$
 A) $\frac{1}{12}$ ☒ B) $\frac{1}{2}$ C) $\frac{1}{3}$ D) $\frac{1}{4}$ E) None
- 4] For $X \sim \text{Poisson}(a)$ with $P(X = 0) = \sqrt{\frac{1}{e}}$, then $P(X > \bar{X}) =$
 A) $\sqrt{\frac{1}{e}}$ ☒ B) $1 - \sqrt{\frac{1}{e}}$ C) $\frac{1}{\sqrt{e}}$ D) $1 - \frac{1}{\sqrt{e}}$ E) None
- 5] Jointly wide-sense stationary orthogonal processes $X(t), Y(t)$ with $R_{XY}(\tau) = 2 + XY$, then $C_{XY}(t, t + \tau) =$
 A) 1 ☒ B) 0 C) 2 D) 3 E) None
- 6] For jointly wide-sense stationary uncorrelated discrete time random processes $X(nT_s), Y(nT_s)$ such that $C_{YY}(nT_s, nT_s + kT_s) = 2 - \bar{X}\bar{Y}$, then $R_{XX}(kT_s) + \bar{X}\bar{Y} =$
 A) 2 B) 4 C) 6 ☒ D) 8 E) None
- 7] If $\text{Var}(X) = 1$, and $Y = X - \bar{X}$, then the correlation of X and Y is
☒ A) 1 B) 2 C) 3 D) 4 E) None
- 8] The pdf of $Y = 2X + 1$, where $X \sim U(0, 1)$, equals
☒ A) 2 B) 25 C) 5 D) 1 E) None
- 9] The variance of the process $X(t) = A$ where $A \sim U(-R, R)$ is equal to
☒ A) $\frac{\pi^2}{3}$ B) $\frac{\pi^3}{3}$ C) $\frac{2\pi^2}{3}$ D) $\frac{2\pi^3}{3}$ E) None
- 10] For $X \sim N(a, a)$, if $\text{std}(X) = E(Y)$, then $\text{std}(Y) =$
 A) 2 ☒ B) 1 C) 6 D) 4 E) None
- 11] If $\alpha > 0$ be a number that satisfies $(-1)^\alpha = -1$. If we have $E(X^2(t)) = t^2 + 2t + \sin(t)$, then $P_{XX} =$
☒ A) 0 B) 3π C) 2π D) π E) None

12] Which of the following is false?

A] The null hypothesis is the one to be tested

B] Null and alternative hypotheses cannot be both true at the same time

C] For $X \sim N(1, 2)$ then $P(X > 1) = 0.5$

D] Type II error occurs when we fail to reject H_0 when it is actually false. ☐

13] Consider the joint PDF $f(x,y)=1$ when $0 < x < 2$, $0 < y < b$ and zero.

A] x

B] $0.5x$

C] $2y$

D] 2

☒ E] None

14] (T/F): For any constant $x > 0$, we have $u(x) > \delta(x)$.

☒ True ☐

15] (T/F): A correlation coefficient $r=0.7$ indicates a strong positive correlation. ☒

16] (T/F): The process $X(t)=A \cos(\omega t)$, $A \sim U(1,2)$ is first order False ☐

17] (T/F): The process $X(t)=\cos(\omega t)$, $\omega \sim U(1,2)$ is first order False ☐

18] (T/F): For some independent random variables X, Y with $E[XY]=E[Y]$, then we must have $E[X]=1$ or $E[Y]=0$. True ☒

19] (T/F): For $X \sim B(n,p)$ if $\frac{\text{Var}(x)}{E[X]} = 2.4$ when $p = \frac{2}{3}$ False ☐

20] (T/F): If $X \sim B(n,0.5)$ and $E[X^2]=5$, then $n=4$.
 $E[X^2] = 0.25n^2 + 0.25n = 5$
True ☒